2022_01 ~ 2022_07

My Activities

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Machine Learning

Part 1-1,

Statistical Model

Statistical Model

Given Project

- To predict whether a machine is functionable based on 21 features.
- Highly imbalanced binary classification.
- Use statistical models for classification as it is structured data.

Statistical Model

Initial Directions

1. Find ways to deal with imbalanced data

2. Find optimal algorithm

3. Find optimal scaler

4. Use cross validation evaluation method for generalization

5. Use f2 score for evaluation as recall is more important in this project.

Statistical Model

Code Developed

1. Find ways to deal with imbalanced data
   - Try oversampling methods ('SMOTE', 'BorderlineSMOTE', 'SVMSMOTE', 'KMeansSMOTE', 'ADASYN', 'SMOTETomek', 'SMOTEENN')

2. Find optimal algorithm
   - Try 2 linear models(logistic regression, ridge), 5 non-linear models(decision tree, KNN, support vector machine, MLP, gaussian process), 3 ensemble models(random forest, XGB, LGBM).

3. Find optimal scaler
   - Try standardize method and normalize method

4. Use cross validation evaluation method for generalization
   - 10fold cross validation

5. Use f2 score for evaluation to put emphasis in recall

Run iterations to find best combination!

Statistical Model

Results

- Used pandas to get whole output into Excel.
- Linear Models with standardize scaling and SMOTE sampling had best result.
- Focus on this combination and add other methods.

LogisticRegression											
Normalize						Standardize					
SMOTE	BorderlineSMOTE	SVMSMOTE	ADASYN	SMOTETomek	SMOTEENN	SMOTE	BorderlineSMOTE	SVMSMOTE	ADASYN	SMOTETomek	SMOTEENN
0.401	0.396	0.356	0.410	0.404	0.401	0.413	0.390	0.327	0.410	0.411	0.418

RidgeClassifier											
Normalize						Standardize					
SMOTE	BorderlineSMOTE	SVMSMOTE	ADASYN	SMOTETomek	SMOTEENN	SMOTE	BorderlineSMOTE	SVMSMOTE	ADASYN	SMOTETomek	SMOTEENN
0.402	0.392	0.341	0.400	0.405	0.409	0.426	0.396	0.334	0.411	0.414	0.400

DecisionTreeClassifier											
Normalize						Standardize					
SMOTE	BorderlineSMOTE	SVMSMOTE	ADASYN	SMOTETomek	SMOTEENN	SMOTE	BorderlineSMOTE	SVMSMOTE	ADASYN	SMOTETomek	SMOTEENN
0.217	0.226	0.229	0.227	0.216	0.279	0.188	0.203	0.201	0.222	0.210	0.296

KNeighborsClassifier											
Normalize						Standardize					
SMOTE	BorderlineSMOTE	SVMSMOTE	ADASYN	SMOTETomek	SMOTEENN	SMOTE	BorderlineSMOTE	SVMSMOTE	ADASYN	SMOTETomek	SMOTEENN
0.318	0.311	0.298	0.312	0.311	0.341	0.336	0.331	0.301	0.344	0.343	0.363

SVC											
Normalize						Standardize					
SMOTE	BorderlineSMOTE	SVMSMOTE	ADASYN	SMOTETomek	SMOTEENN	SMOTE	BorderlineSMOTE	SVMSMOTE	ADASYN	SMOTETomek	SMOTEENN
0.323	0.319	0.315	0.315	0.309	0.338	0.279	0.281	0.259	0.278	0.290	0.353

MLPClassifier											
Normalize						Standardize					
SMOTE	BorderlineSMOTE	SVMSMOTE	ADASYN	SMOTETomek	SMOTEENN	SMOTE	BorderlineSMOTE	SVMSMOTE	ADASYN	SMOTETomek	SMOTEENN
0.349	0.337	0.323	0.348	0.339	0.364	0.265	0.269	0.247	0.270	0.249	0.358

GaussianProcessClassifier											
Normalize						Standardize					
SMOTE	BorderlineSMOTE	SVMSMOTE	ADASYN	SMOTETomek	SMOTEENN	SMOTE	BorderlineSMOTE	SVMSMOTE	ADASYN	SMOTETomek	SMOTEENN
0.371	0.378	0.345	0.385	0.380	0.382	0.277	0.253	0.234	0.281	0.279	0.363

RandomForestClassifier											
Normalize						Standardize					
SMOTE	BorderlineSMOTE	SVMSMOTE	ADASYN	SMOTETomek	SMOTEENN	SMOTE	BorderlineSMOTE	SVMSMOTE	ADASYN	SMOTETomek	SMOTEENN
0.202	0.192	0.146	0.199	0.200	0.302	0.161	0.155	0.130	0.148	0.171	0.321

XGBClassifier											
Normalize						Standardize					
SMOTE	BorderlineSMOTE	SVMSMOTE	ADASYN	SMOTETomek	SMOTEENN	SMOTE	BorderlineSMOTE	SVMSMOTE	ADASYN	SMOTETomek	SMOTEENN
0.190	0.197	0.190	0.180	0.194	0.338	0.168	0.176	0.174	0.171	0.174	0.348

LGBMClassifier											
Normalize						Standardize					
SMOTE	BorderlineSMOTE	SVMSMOTE	ADASYN	SMOTETomek	SMOTEENN	SMOTE	BorderlineSMOTE	SVMSMOTE	ADASYN	SMOTETomek	SMOTEENN
0.178	0.183	0.169	0.171	0.156	0.315	0.138	0.126	0.139	0.147	0.152	0.329

Statistical Model

Modification Directions

- Linear models assume linear independency. Try chi-square independency test and check correlation
  - Create feature combinations only with elements with linear independency in between.

- Try IQR for outlier detection.
  - Remove outliers

- Try one-hot encoding to remove the effect of class label number.

Statistical Model

Modification Results

- Effect of Independency Test

- Using independency test made average scores to be slightly lower than control group. (fig1)

- However, using independency test have possibility in achieving higher score than control group in certain combination. (fig2)

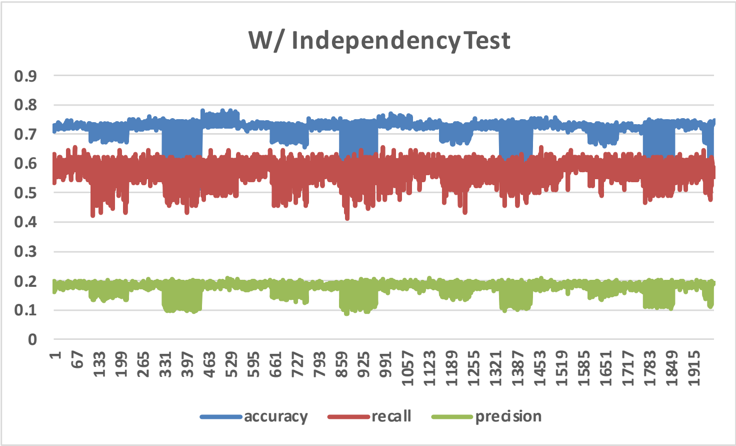
W/O IndependencyTest

Accuracy	0.7330
Recall	0.5778
Precision	0.1839

W/ I.Test

Average Accuracy	0.713659
Average Recall	0.573474
Average Precision	0.176252

↑ Fig1. Average Scores



↑ Fig2. Score plot of different combination of independent features

Statistical Model

Modification Results

- Effect of IQR outlier removal

- Using IQR outlier removal had negative effect.

W/O IQR

Average Accuracy	0.713659
Average Recall	0.573474
Average Precision	0.176252

MAX Accuracy	Accuracy	0.781
	Recall	0.489
	Precision	0.200
MAX Recall	Accuracy	0.737
	Recall	0.656
	Precision	0.205
MAX Precision	Accuracy	0.751
	Recall	0.611
	Precision	0.209

W/ IQR

Average Accuracy	0.708721
Average Recall	0.571053
Average Precision	0.171034

MAX Accuracy	Accuracy	0.787
	Recall	0.488
	Precision	0.193
MAX Recall	Accuracy	0.668
	Recall	0.513
	Precision	0.129
MAX Precision	Accuracy	0.734
	Recall	0.663
	Precision	0.212

Statistical Model

Modification Results

- Effect of one-hot encoding

- Using one-hot encoding improved the overall result.

W/O One-Hot

Average Accuracy	0.693177
Average Recall	0.54421
Average Precision	0.158778

W/ One-Hot

Average Accuracy	0.729171
Average Recall	0.600303
Average Precision	0.188493

Part 1-2,

CNN
Model

CNN Model

Tutorial Project - MNIST

- Create CNN architecture based on LeNET5.
  - Changed # of filters, added 1 layer, added batch normalization.

- Use data augmentation methods.
  - Used image data generator from keras.

- Use 10 model ensemble method to achieve higher accuracy.

CNN Model

Tutorial Project - MNIST

Architecture

Modification

Process

Base Model (LeNet5): Conv2D(3x3 filter 6) - Max Pool - Conv2D(5x5 filter 16) - Flatten - Dense 120 - Dense 84

* Red: improved performance (add), Orange: decreased performance (delete)

Modification 1:
Conv2D(3x3 filter 32) - Max Pool - Conv2D(5x5 filter 64) - Max Pool - Flatten - Dense 120 - Dense 84

Modification 2:
Conv2D(3x3 filter 32) - Max Pool - Conv2D(5x5 filter 64) - Max Pool - Conv2D(5x5 filter 128) - Max Pool - Flatten - Dense 120 - Dense 84

Modification 3:
Conv2D(3x3 filter 32) - Max Pool - Conv2D(5x5 filter 64) - Max Pool - Conv2D(5x5 filter 128) - Max Pool - Conv2D(5x5 filter 128) - Max Pool - Flatten - Dense 120 - Dense 84

Modification 4:
Conv2D(3x3 filter 32) - Max Pool - Conv2D(5x5 filter 64) - Max Pool - Conv2D(5x5 filter 128) - Max Pool - Flatten - Dense 120 ~~- Dense 84~~

Modification 5:
Conv2D(3x3 filter 32) - Max Pool - Conv2D(3x3 filter 32) - Max Pool - Conv2D(5x5 filter 64) - Max Pool - Conv2D(5x5 filter 128) - Max Pool - Flatten - Dense 120 ~~- Dense 84~~

Modification 6:
Conv2D(3x3 filter 32) -Batch Normalization- Max Pool - Conv2D(3x3 filter 32) -Batch Normalization- Max Pool - Conv2D(5x5 filter 64) - Batch Normalization- Max Pool - Conv2D(5x5 filter 128) -Batch Normalization- Max Pool - Flatten - Dense 120

Modification 7:
Conv2D(3x3 filter 32) -Batch Normalization- Max Pool - Conv2D(3x3 filter 32) -Batch Normalization- Max Pool - Conv2D(5x5 filter 64) - Batch Normalization- Max Pool - Conv2D(5x5 filter 128) -Batch Normalization- Max Pool - Flatten - Dense 120 - Dropout(0.5)

Modification 8: Added Data Augmentation & 10 model ensemble

Final Model:
Conv2D(3x3 filter 32) -Batch Normalization- Max Pool - Conv2D(3x3 filter 32) -Batch Normalization- Max Pool - Conv2D(5x5 filter 64) -Batch Normalization- Max Pool - Conv2D(5x5 filter 128) -Batch Normalization- Max Pool - Flatten - Dense 120

Final Model Result

Test 1

CNN 1: Train accuracy=0.99291, Validation accuracy=0.99483, Test accuracy=0.99470
CNN 2: Train accuracy=0.99376, Validation accuracy=0.99400, Test accuracy=0.99230
CNN 3: Train accuracy=0.99261, Validation accuracy=0.99400, Test accuracy=0.99410
CNN 4: Train accuracy=0.99389, Validation accuracy=0.99400, Test accuracy=0.99390
CNN 5: Train accuracy=0.99378, Validation accuracy=0.99483, Test accuracy=0.99420
CNN 6: Train accuracy=0.99467, Validation accuracy=0.99433, Test accuracy=0.99370
CNN 7: Train accuracy=0.99478, Validation accuracy=0.99433, Test accuracy=0.99470
CNN 8: Train accuracy=0.99480, Validation accuracy=0.99450, Test accuracy=0.99450
CNN 9: Train accuracy=0.99456, Validation accuracy=0.99567, Test accuracy=0.99430
CNN 10: Train accuracy=0.99491, Validation accuracy=0.99417, Test accuracy=0.99380
==

Final Ensembled Score : Accuracy = 0.99740

Test 2

CNN 1: Train accuracy=0.99409, Validation accuracy=0.99467, Test accuracy=0.99440
CNN 2: Train accuracy=0.99474, Validation accuracy=0.99417, Test accuracy=0.99490
CNN 3: Train accuracy=0.99387, Validation accuracy=0.99383, Test accuracy=0.99470
CNN 4: Train accuracy=0.99556, Validation accuracy=0.99450, Test accuracy=0.99510
CNN 5: Train accuracy=0.99444, Validation accuracy=0.99600, Test accuracy=0.99460
CNN 6: Train accuracy=0.99317, Validation accuracy=0.99317, Test accuracy=0.99380
CNN 7: Train accuracy=0.99357, Validation accuracy=0.99367, Test accuracy=0.99430
CNN 8: Train accuracy=0.99333, Validation accuracy=0.99333, Test accuracy=0.99330
CNN 9: Train accuracy=0.99485, Validation accuracy=0.99483, Test accuracy=0.99420
CNN 10: Train accuracy=0.99269, Validation accuracy=0.99333, Test accuracy=0.99180
==

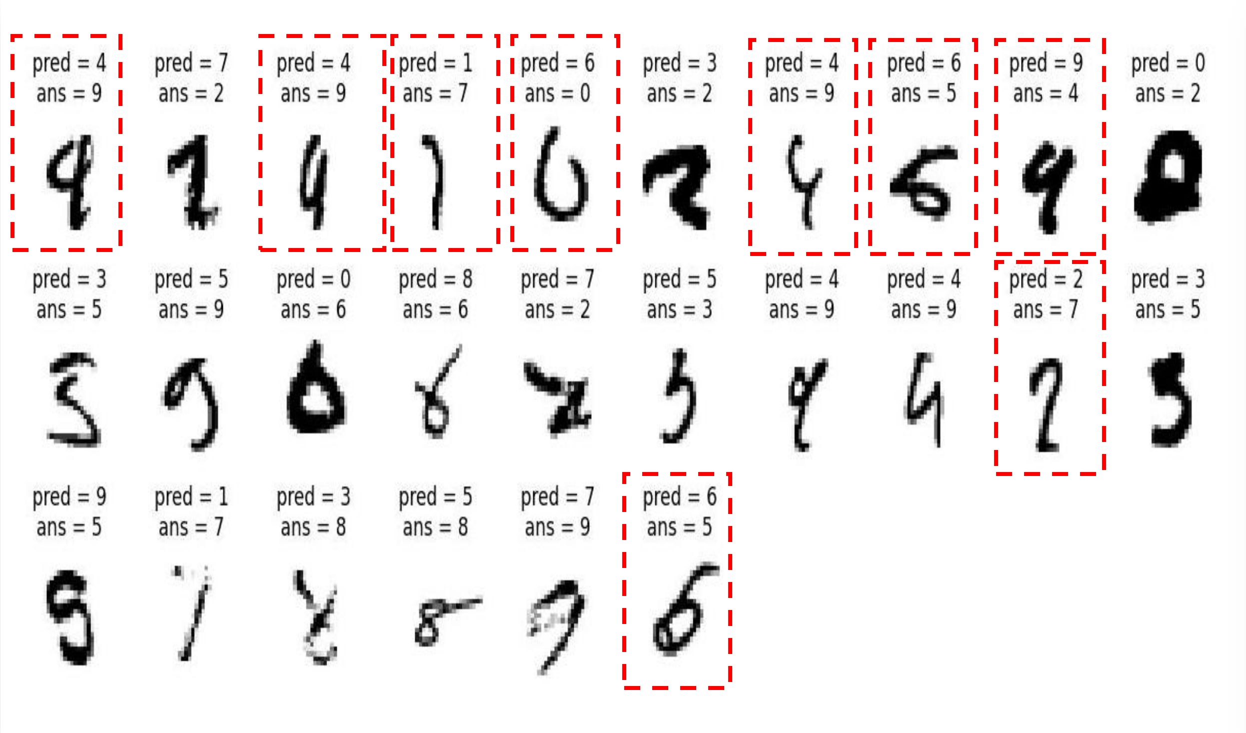
Final Ensembled Score : Accuracy = 0.99700

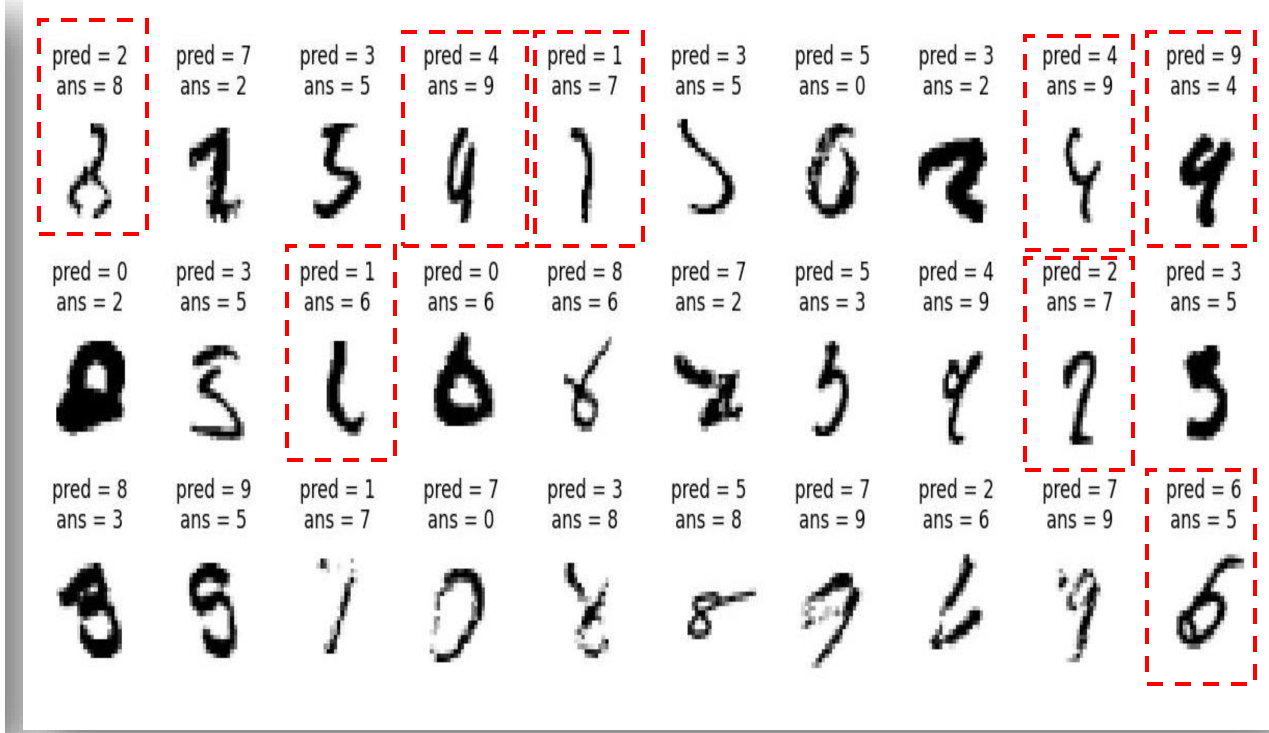
CNN Model

Misclassified Images

Test 1

Test 2



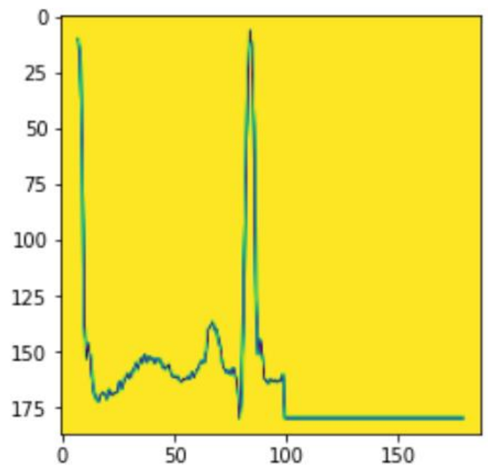


Part 1-2

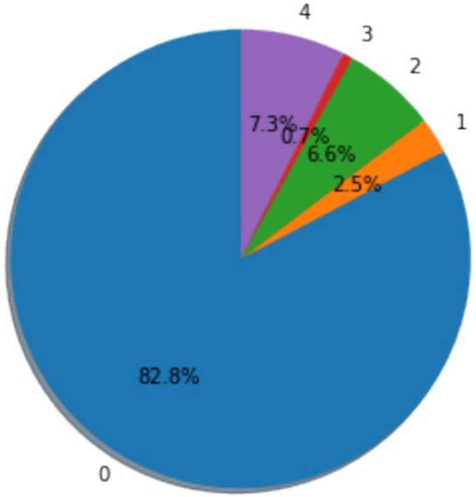
CNN Model

Project 2: MIT-BIH Arrhythmia Classification

- MIT-BIH dataset : A dataset that labeled ECG into normal, 3 types of arrhythmia and unknown.
- Used final model from MNIST project as base model.
- Used cost-sensitive method to deal with imbalanced data.



↑ Example of dataset
(Label : Normal)



↑ Class distribution
of dataset (Highly imbalanced)

CNN Model

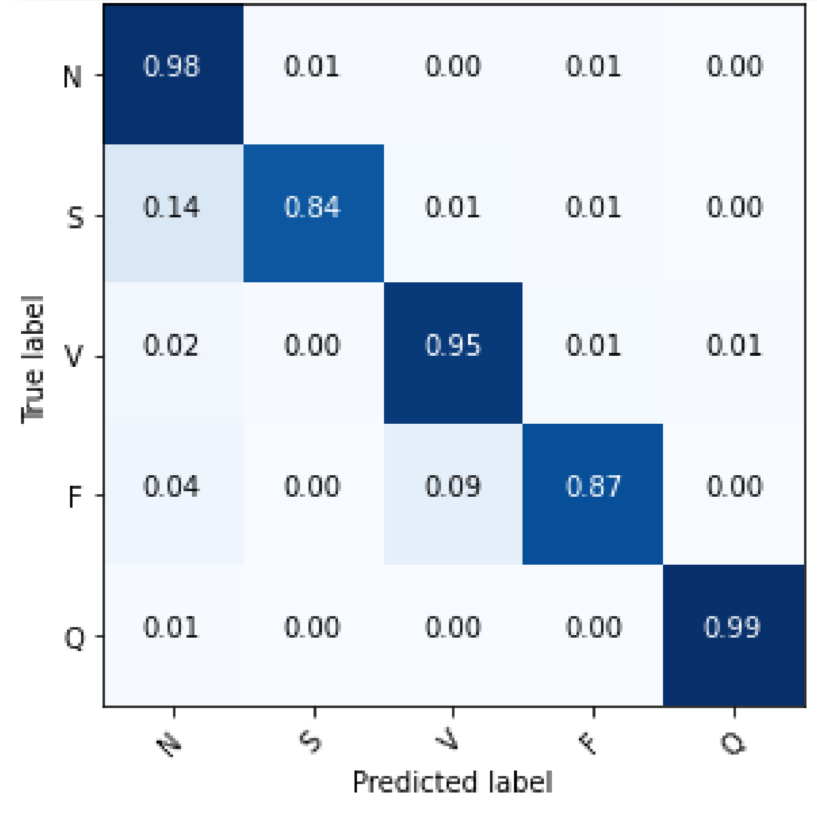
Base Model

TEST DATA	Accuracy	Precision	Recall	F1-score
0 (Normal)	0.972	0.99	0.98	0.98
1 (Atrial Premature Beat)		**0.73**	**0.84**	**0.78**
2 (Premature Ventricular Contraction)		0.93	0.95	0.94
3 (Fusion Beat)		**0.45**	**0.87**	**0.60**
4 (Unknown)		0.97	0.99	0.98

- Showed great performance in predicting class 0, 2 and 4. Focused on improving performance for class 1 and 3.
- Precision of class 3 was significantly low (0.45) compared to other scores.

CNN Model

Base Model



- Class 1 (S) is being confused with class 0 (N) the most.
- Class 3 (F) is being confused with class 0 (N) and class 2 (V) the most.

CNN Model

Train / Validation Scores

VAL DATA	Accuracy	Precision	Recall	F1-score
0	0.975	0.99	0.98	0.98
1		0.73	0.83	0.78
2		0.93	0.95	0.94
3		0.44	0.87	0.58
4		0.97	0.99	0.98
TRAIN DATA	**Accuracy**	**Precision**	**Recall**	**F1-score**
0	0.983	1.00	0.98	0.99
1		0.81	0.97	0.88
2		0.96	0.99	0.97
3		0.49	0.99	0.66
4		0.98	1.00	0.99

- Looking at the classification performance of validation data, validation data and test data had no noticeable difference in score, so I could confirm that the data split was well performed.

- Looking at the classification performance of train data, the model performed with higher score overall compared to validation or test sets. However, the precision of class 3 was significantly low (0.49) even in the train set.

Modification Direction

1. Modify Weights

   Model is insensitive in predicting class 1 (low recall) and too sensitive in class 3 (low precision). Try increasing the weight of class 1 and decreasing that of class 3.

2. Deeper Model

   The sensitivity may be caused by model's depth not being enough. Especially, the precision score of class 3 in train data is low as well (0.49), so it supports the idea that the model may be not deep enough. Try making it deeper.

Result1 – Modify Weights

Modify Weights	Acc	Pre	Rec	F1
0	0.971	0.99	0.98	0.98
1		**0.71**	**0.82**	**0.76**
2		0.91	0.95	0.93
3		**0.55**	**0.88**	**0.67**
4		0.96	0.99	0.98

↑ Fig1. performance of weight changed model

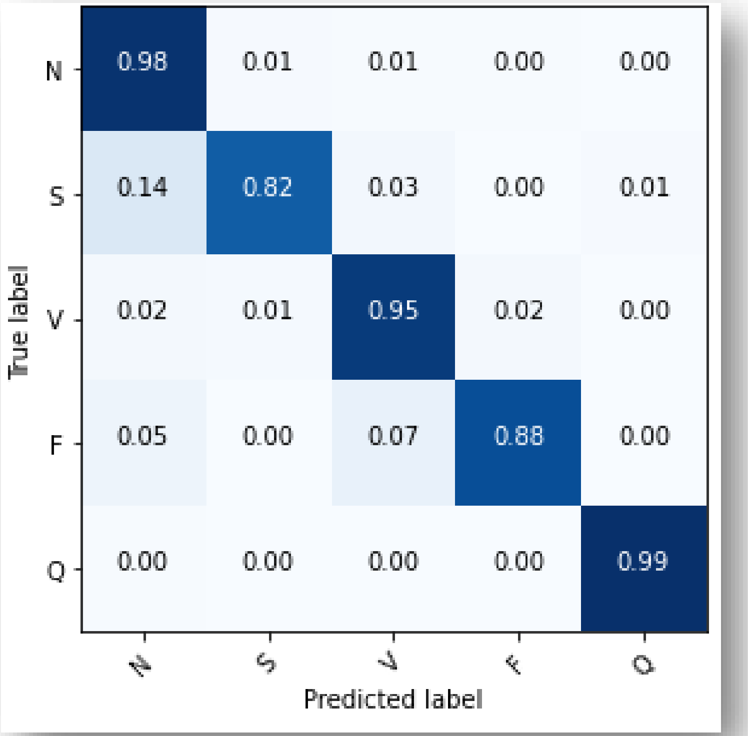
Base Model	Acc	Pre	Rec	F1
0	0.972	0.99	0.98	0.98
1		**0.73**	**0.84**	**0.78**
2		0.93	0.95	0.94
3		**0.45**	**0.87**	**0.60**
4		0.97	0.99	0.98

↑ Fig2. performance of base model

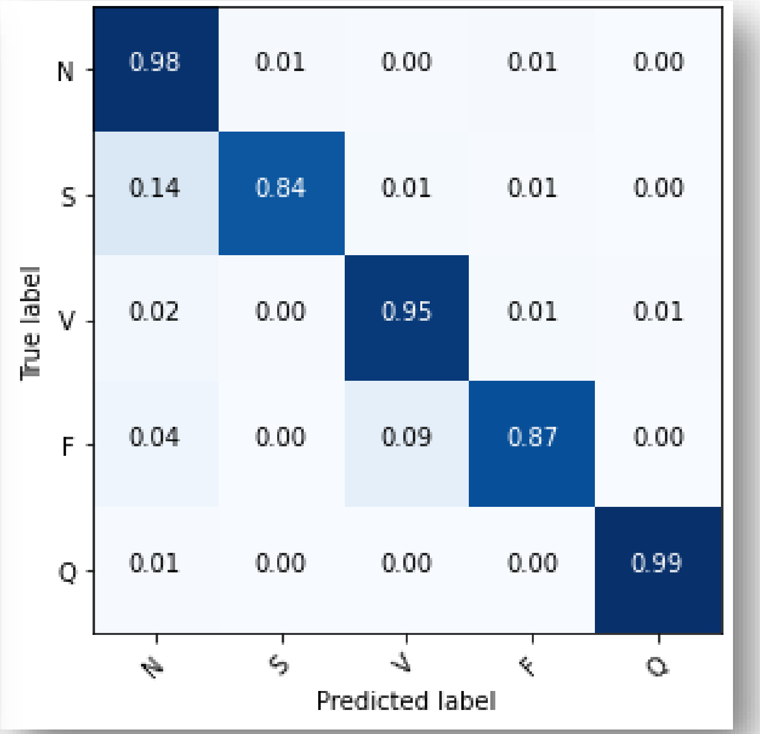
- The prediction performance of class 1 decreased.
- The performance increased for class 3. (accuracy increased by 1%, precision by 10%, and recall by 1%).

CNN Model

Result1 – Modify Weights



↑ Fig1. Confusion Matrix of Modified Weight Model



↑ Fig2. Confusion Matrix of Base Model

- Changed weight of class 1, and the classification performance between class 1 and 0 stayed the same. Further, the classification performance between class 1 and 2 decreased.

- Class 3 didn't have noticeable difference in confusion matrix.

Result1 – Modify Weights

Conclusion

- Keep the modified weight of class 3.
- Use the original weight for class 1.

Original Weights	{0: 1, 1: 32.601, 2: 12.521, 3: 113.059, 4: 11.269}
Modified Weights	{0: 1, 1: 50, 2: 12.521, 3: 100, 4: 11.269}
Final Weight	{0: 1, 1: 32.601, 2: 12.521, 3: 100, 4: 11.269}

Result2 – Deeper Model

Deeper Model	Acc	Pre	Rec	F1
0	0.980	0.99	0.99	0.99
1		**0.80**	**0.81**	**0.80**
2		0.94	0.96	0.95
3		**0.63**	**0.88**	**0.74**
4		0.99	0.99	0.99

↑ Fig1. performance of deeper model

Base Model	Acc	Pre	Rec	F1
0	0.972	0.99	0.98	0.98
1		**0.73**	**0.84**	**0.78**
2		0.93	0.95	0.94
3		**0.45**	**0.87**	**0.60**
4		0.97	0.99	0.98

↑ Fig2. performance of base model

Modify Weights	Acc	Pre	Rec	F1
0	0.971	0.99	0.98	0.98
1		**0.71**	**0.82**	**0.76**
2		0.91	0.95	0.93
3		**0.55**	**0.88**	**0.67**
4		0.96	0.99	0.98

↑ Fig3. performance of weight modified model

- Added 1 convolutional layer with 128 filters.
- Deeper Model(fig1) vs. Original Model(fig2)

  Increasing depth had positive effect in the classification of class 3.

  Class1: accuracy and recall decreased 3%, precision increased 7%.

  Class3: accuracy and recall increased 1%, precision increased 18%.
- Deeper Model(fig1) vs. Modified Weights(fig3)

  Making model deeper had a bigger effect on improving the classification performance of class 1 and 3.

Result2 – Deeper Model

Conclusion

- Increasing depth while using the modified weight of class 3 expected to have positive effect.
- Try deeper models.

Result3 – Modified Weights + Deeper Model

Modified Weight +Deeper Model	Acc	Pre	Rec	F1
0	0.981	0.99	0.99	0.99
1		**0.82**	**0.83**	**0.82**
2		0.95	0.95	0.95
3		**0.68**	**0.86**	**0.76**
4		0.98	0.99	0.99

↑ Fig1. performance of model with both weight and depth modification

Modify Weights	Acc	Pre	Rec	F1
0	0.975	0.99	0.98	0.98
1		**0.71**	**0.82**	**0.76**
2		0.91	0.95	0.93
3		**0.55**	**0.88**	**0.67**
4		0.96	0.99	0.98

↑ Fig2. performance of weight modified model

Deeper Model	Acc	Pre	Rec	F1
0	0.975	0.99	0.98	0.98
1		**0.71**	**0.82**	**0.76**
2		0.91	0.95	0.93
3		**0.55**	**0.88**	**0.67**
4		0.96	0.99	0.98

↑ Fig3. performance of depth modified model

- Modified Weights + Deeper(fig1) vs. Weight Modified(fig2)

  Modifying weight and depth together made the performance increase overall.

- Modified Weights + Deeper(fig1) vs. Depth Modified(fig3)

  Modifying weight and depth together made the performance increase overall.

- Modifying weights and depth together had positive effect.

Result4 – Try Extra Deeper Model

Weight + Extra Deeper	Acc	Pre	Rec	F1
0	0.977	0.99	0.98	0.99
1		**0.82**	**0.82**	**0.82**
2		0.95	0.94	0.94
3		**0.48**	**0.91**	**0.63**
4		0.99	0.99	0.99

↑ Fig1. performance of model with both weight and extra depth modification

Weight + Deeper	Acc	Pre	Rec	F1
0	0.981	0.99	0.99	0.99
1		**0.82**	**0.83**	**0.82**
2		0.95	0.95	0.95
3		**0.68**	**0.86**	**0.76**
4		0.98	0.99	0.99

↑ Fig2. performance of model with both weight and depth modification

- Added one more convolutional layer.
- There was no difference overall but had fatal decrease in performance of class 3 classification. (precision decreased 20% in class3)
- Extra Depth wasn't a good idea.

Result5 – Modified Weights + Deeper Model + Ensemble

Ensembled (Final Model)	Acc	Pre	Rec	F1
0	0.986	0.99	0.99	0.99
1		**0.90**	**0.84**	**0.87**
2		0.96	0.96	0.96
3		**0.71**	**0.88**	**0.79**
4		0.99	1.00	0.99

↑ Fig2. performance of ensembled model with both weight and depth modification

Modified Weight +Deeper Model	Acc	Pre	Rec	F1
0	0.981	0.99	0.99	0.99
1		**0.82**	**0.83**	**0.82**
2		0.95	0.95	0.95
3		**0.68**	**0.86**	**0.76**
4		0.98	0.99	0.99

↑ Fig1. performance of model with both weight and depth modification

- Ensembled 10 model with modified weight and depth.
- In class 1, precision increased 8%, and recall increased 1%.
- In class 3, precision increased 3%, and recall increased 2%.
- The performance in class 2 and 4 increased as well.

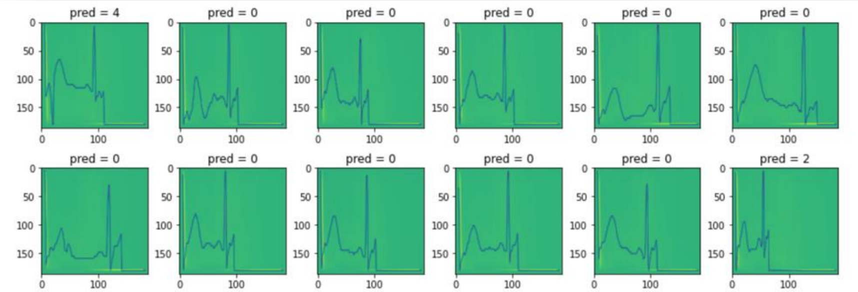
Toy project : Application on Apple Watch ECG

↑ Apple Watch ECG report

↑ Fragment from whole report

- Transformed my ECG report from apple watch to a test data and checked if my model can be used for actual application.

- Got output like the figure below.



← Output from my ECG

CFD Projects
-Artificial Blood Vessels

Part 2-1,

Vascular Bypass Surgery

Vascular Bypass Surgery

Introduction

- Atherosclerosis is a leading cause of death around the world, a disease caused by the deposition of plaques in inner walls of arteries.
- In severe cases of atherosclerosis, invasive procedures such as vascul ar bypass surgery is required.
- To achieve higher patency in the procedure, it is important to control the wall shear stress produced by blood flow to be above a certain l evel, as it was previously discovered that low wall shear stress is signi ficantly correlated with atherosclerosis.
- Aim : to understand the correlation between anastomosis angle and wall shear stress distributions in a totally occluded vessel.

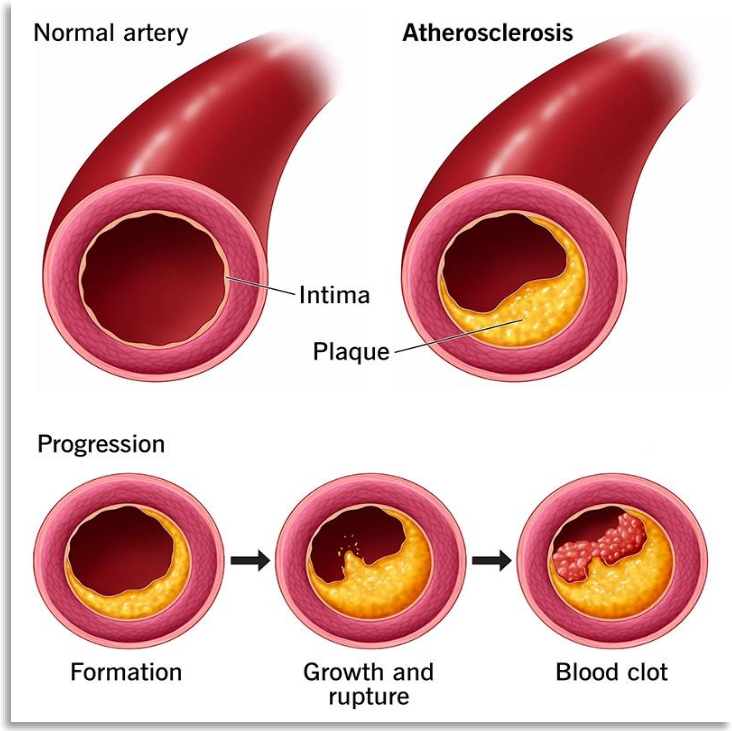


Figure 1. Development of Atherosclerosis

Image source : Cleveland Clinic

Vascular Bypass Surgery

Numerical Model

- Produced six simplified models of bypassed blood vessel to view the velocity distribution and the following wall shear stress according to anastomosis angle.
- The diameter of host vessel (D_h) was set to 4mm and the bypass graft thickness (D_g) was set to $0.9D_h$. (figure 1)
- The proximal host vessel length (l_p), distal host vessel length (l_d) and bypass le ngth were all set to $10D_h$.
- The anastomosis angles (α) were set to 15°, 30°, 45°, 60°, 75°, and 90° respectiv ely, for models A, B, C, D, E and F (figure 2).

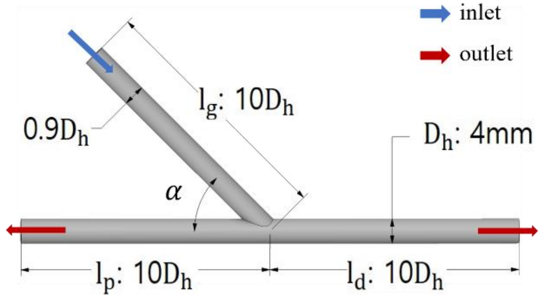


Figure 1. Common dimensions of simplified models. Anastomosis angle (α) were used as only variable.

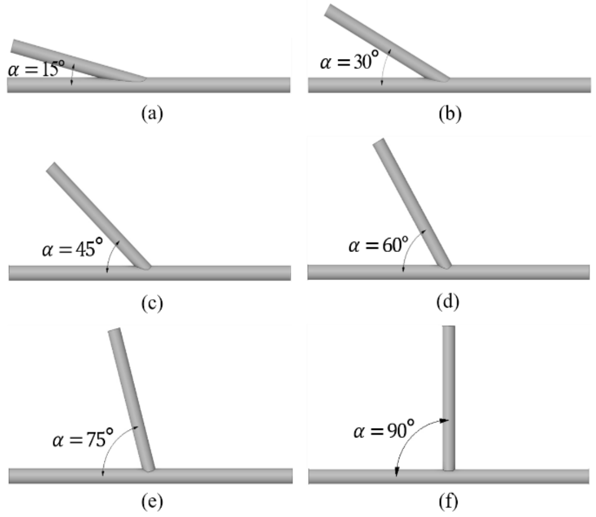


Figure 2. Anastomosis angles in model A~F
a) model A (α = 15°). b) model B (α = 30°). c) model C (α = 45°). d) model D (α = 60°). e) model E (α = 75°). f) model F (α = 90°).

Vascular Bypass Surgery

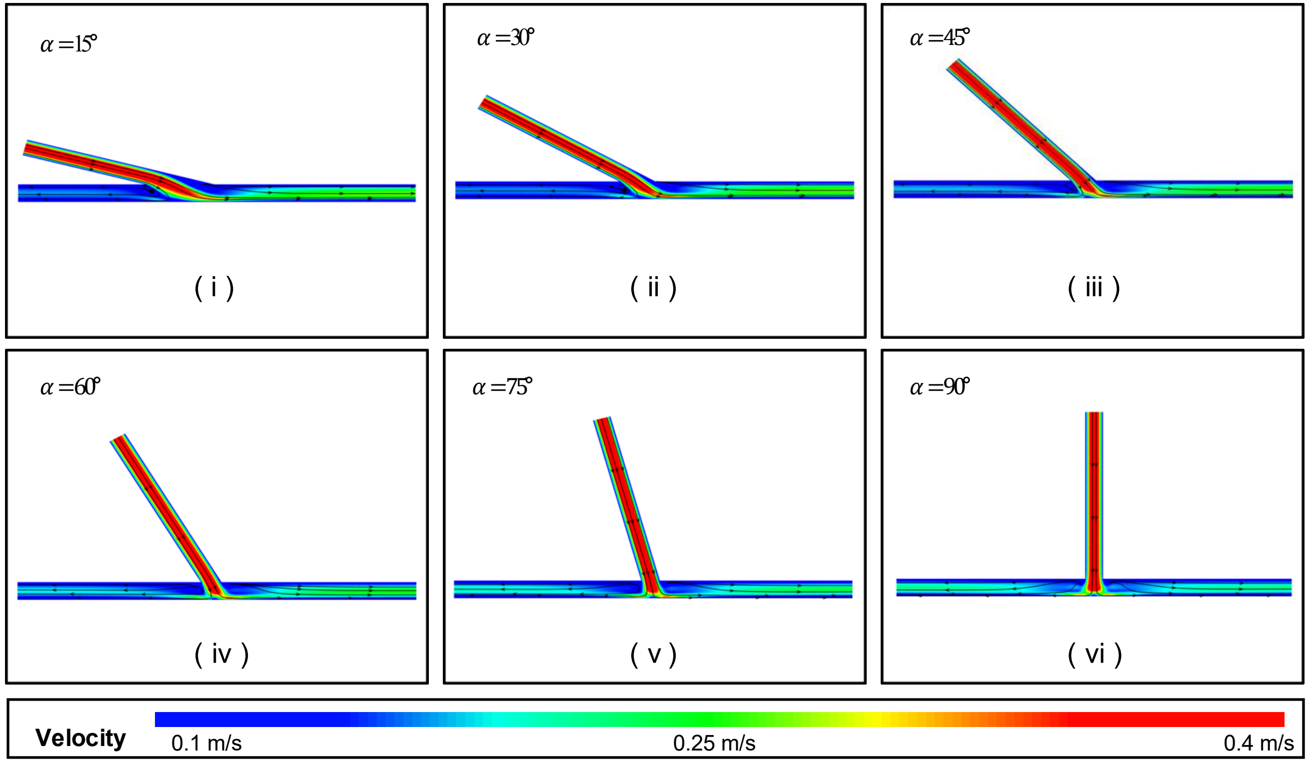
Numerical Method

- Assumptions:
    - Blood flow was assumed to be laminar.
    - The effects of cardiac movements (i.e. contraction and relaxation) were neglected, the flow was assumed to be steady.
    - Blood was assumed to be incompressible Newtonian fluid.

- Numerical simulation:
    - Finite volume method was used to solve the governing equations: continuity equations and Navier-Stokes equations.
    - Inlet boundary conditions as fully-developed laminar flow with Reynolds number (Re) of 250.
    - Neumann outlet conditions, and pressure to be zero at both outlets.

Vascular Bypass Surgery

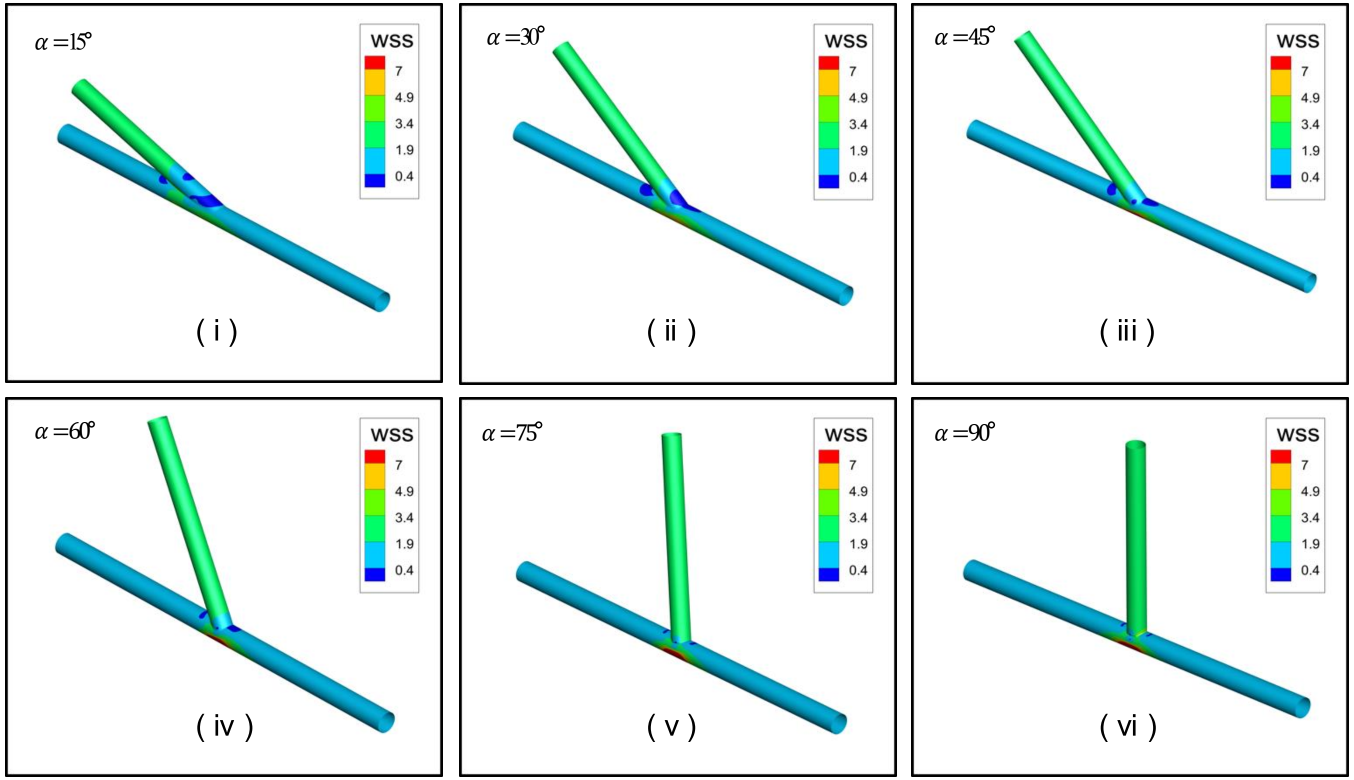
Result - Velocity Contour & Streamlines



- Flow with low velocity was observed in graft anastomosis in model i and ii (15° and 30°)
- Velocity in proximal area was faster in models with anastomosis angle above 45°.
- Anastomosis angle had no effect on high velocity flow (above 0.3m/s).
- Recirculating flow was formed in proximal anastomosis area in models with anastomosis angle below 45°.

Vascular Bypass Surgery

Result – Wall Shear Stress Contour

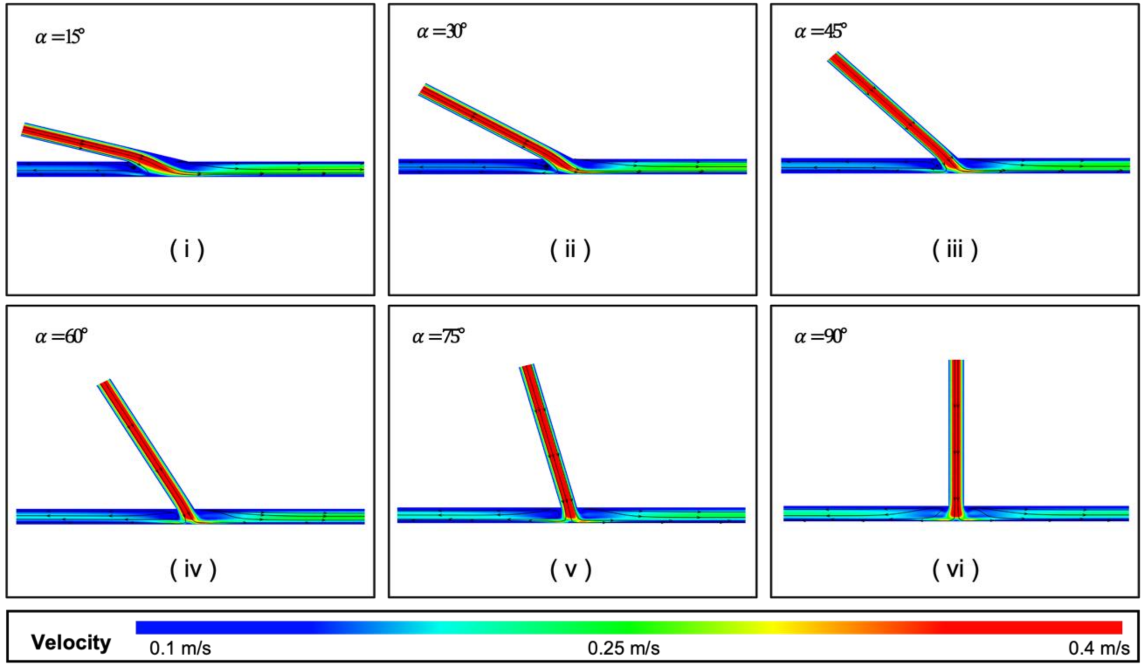


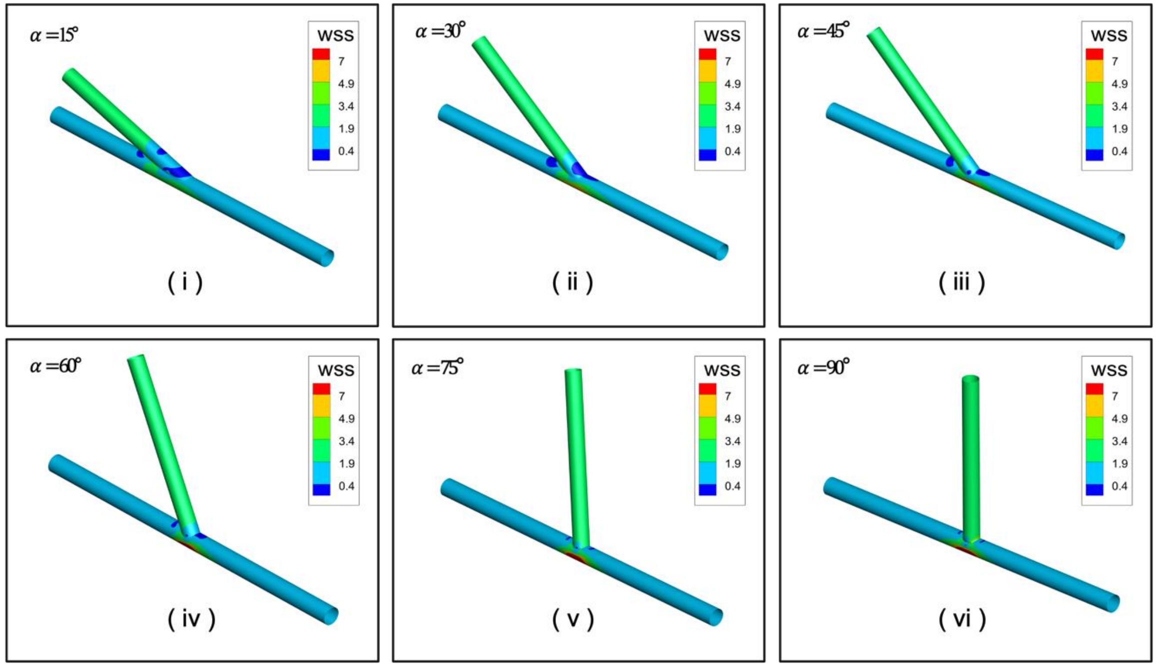
Result

- Low wall shear stress was concentrated near anastomosis in all models.
- The size of low wall shear stress region decreased as anastomosis angle increased.

Vascular Bypass Surgery

Conclusion





- As the size of low wall shear stress region decreases near anastomosis, increasing anastomosis angle may reduce the risk of restenosis near anastomosis.

Part 2-1

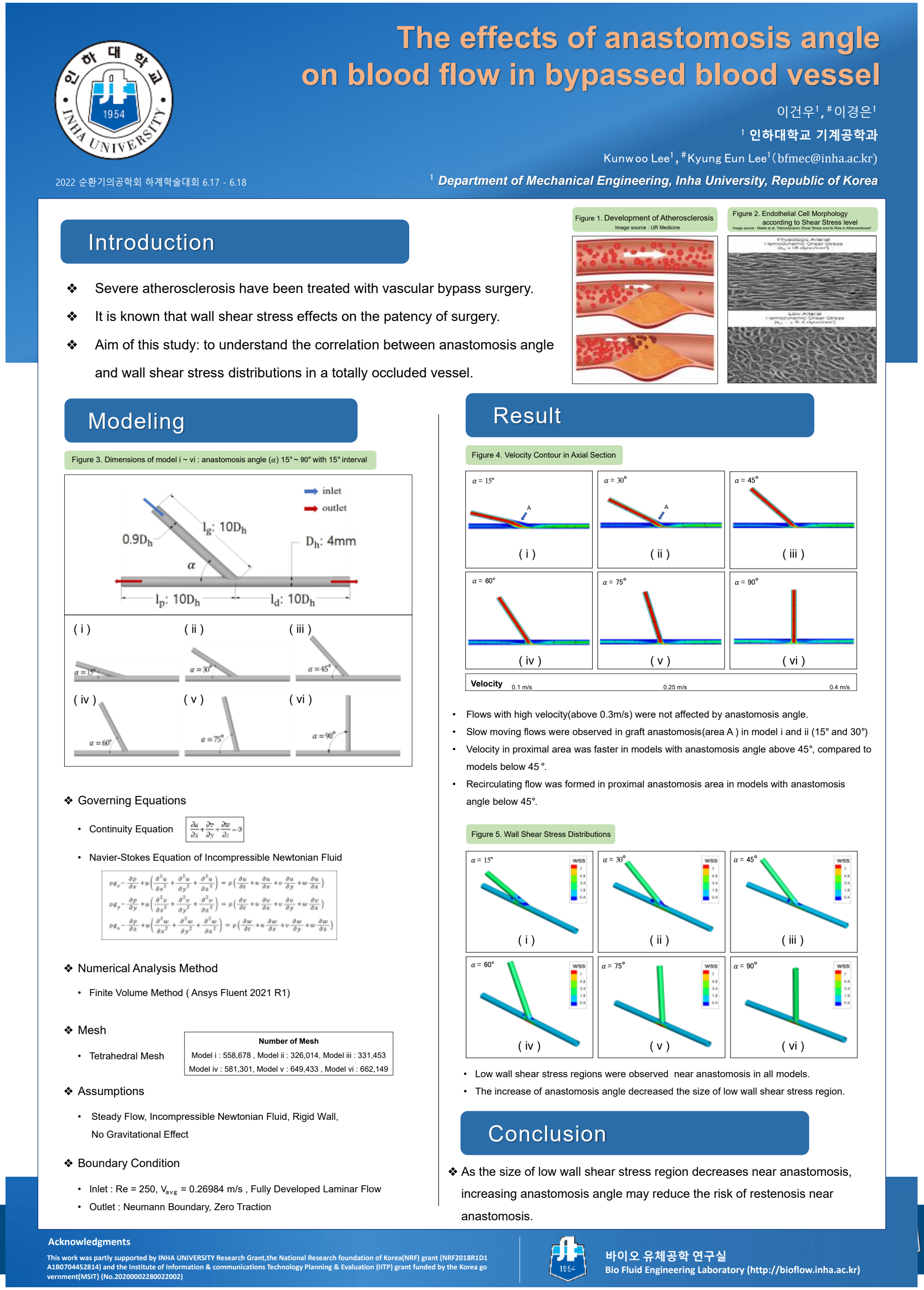
Vascular Bypass Surgery

*Poster for the project

Made poster of the project for the

summer conference 2022 of

Biomedical Engineering Society for Circulation



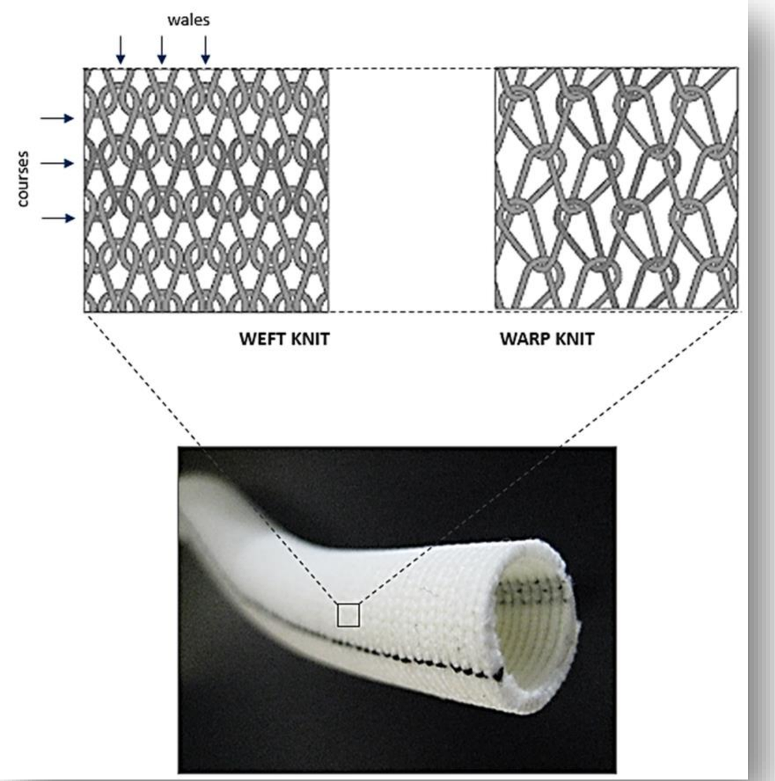
Part 2-2,

Artificial Blood Vessel w/ Pattern

Introduction

Synthetic Non-Degradable Polymers

- Dacron, Teflon(ePTFE) or Polyurethane is used for synthetic non-degradable polymer artificial vessel.
- Healthy autologous vessel has limit in its supply, raising the need for polymer artificial vessel.
- Synthetic Non-degradable Polymers is widely used as it has advantage in the ease at changing mechanical properties.



↑ Example of Dacron artificial vessel.

Challenges of Non-Degradable Polymers

- Poor Patency Rates

  ex) Only 45% of standard ePTFE grafts are patent at 5 years as femoropopliteal bypass grafts

  ↔ 60-80% in autologous vein grafts. [1]

- Compliance Mismatch

  The compliance of polymer may be dozens of times bigger than soft tissues in our body, so leaving the compliance mismatch may cause problems in healing process or cause intimal hyperplasia. [2]

- Enhancing biocompatibility of synthetic material

  Low biocompatibility cause inflammatory responses. We can enhance biocompatibility via surface modification, coatings, chemical and protein modifications, endothelial cell seeding …

Advantages of Porous Synthetic Blood Vessel

1. Positive effect on biocompatibility
   - Endothelial cell has a shape with certain direction. Thus, having similar pattern in inner wall can have positive effect on endothelialization, increasing patency rate. [3]
   - Putting substances such as nitric oxide(NO), dopamine, heparin may help control inflammatory responses, inhibiting the formation of thrombus. [4]

2. Help solve compliance mismatch
   - Having porosity can help match human body's compliance.

Hemodynamic Factors and Patency

- Having porosity may have previously discussed pros, but flow disturbance due to porosity may cause side effects in hemodynamic aspects.

- Low wall shear stress may cause restenosis by intimal thickening

  In our body, blood vessels react to low shear stress by contraction, causing the blood velocity to temporally rise. Synthetic blood vessels can't react the same way, causing the endothelial cell in the area produce certain vasoactive substance, which leads to intimal thickening. [5]

- Recirculation may be formed.

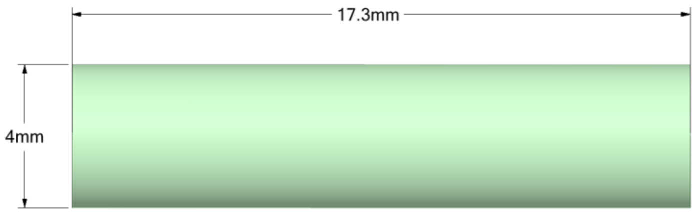
Aim of the Study

Model porous artificial blood vessel with a simplified model with pattern on the inner wall and investigate the effect of porosity on patency rate in the aspect of hemodynamics.

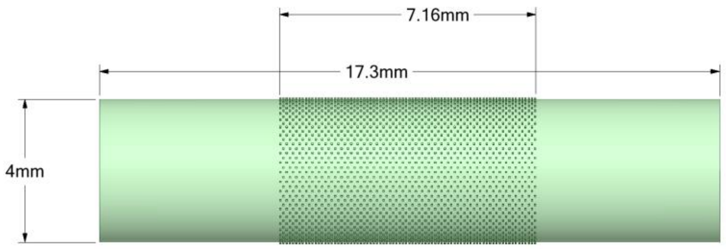
Modeling

Models

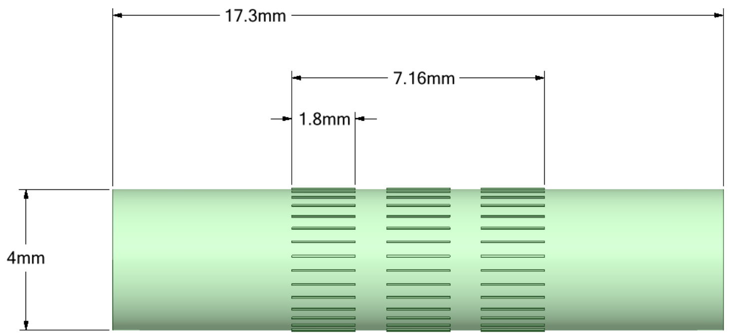
- Produced 3 models. Model I (No pattern), Model II (Cubic shaped pattern), ModelI III (Cuboid shaped pattern)
- The cube pattern of Model II has a side of 45 μm, and 47 pieces in the circumferential direction and 80 pieces in the axial direction are attached to the surface of a cylinder with a diameter of 4mm and a length of 7.16mm. For stability of analysis, cylinders were added to both ends forming a model with a total length of 17.3 mm.
- The cuboid pattern of Model III has the shape of the cube of Model II stretched 40 times in the axial direction. 31 cuboids are in the circumferential direction and 3 cuboids are in the axial direction, attached to the surface of a 7.16mm long cylinder. As in Model II, cylinders were added at both ends for stability.



↑ Model I



↑ Model II



↑ Model III

Modeling

Method

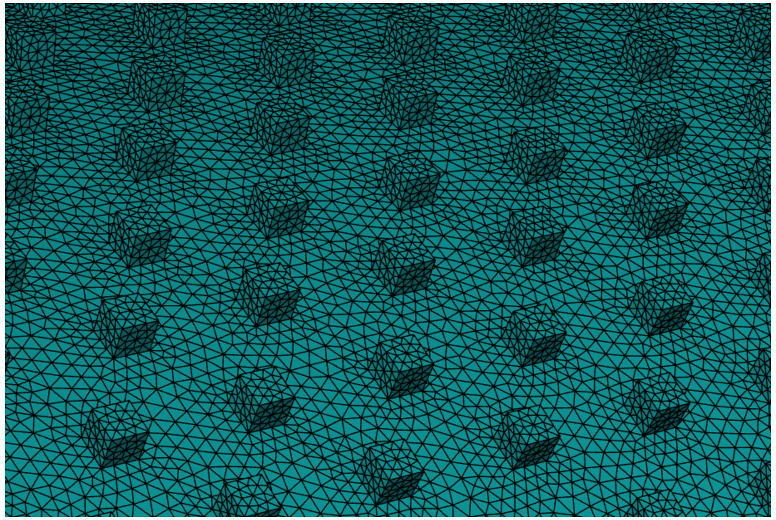
- Governing Equation : Continuity Eq. & Navier-Stokes Eq.
- Finite Volume Method (Fluent v.2021 R1)
- Tetrahedral Mesh

  Number of Mesh - Model I : 1,025,564 / Model II : 11,311,373 / Model III : 6,071,591
- Assumptions

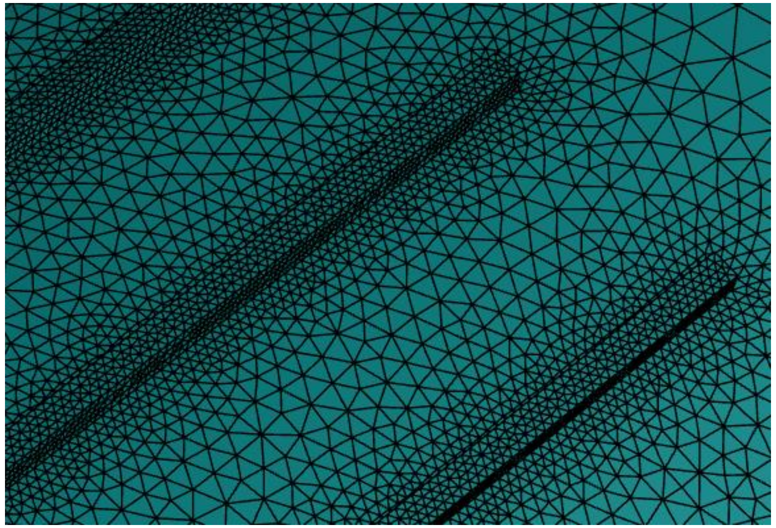
  Steady Flow, Incompressible Newtonian Fluid, Rigid Wall, No Gravitational Effect
- Boundary Condition

  Inlet : $\mathrm{Re} = 250$, $V_{avg} = 0.242857\ m/s$, Fully Developed Laminar Flow

  Outlet : Neumann Boundary



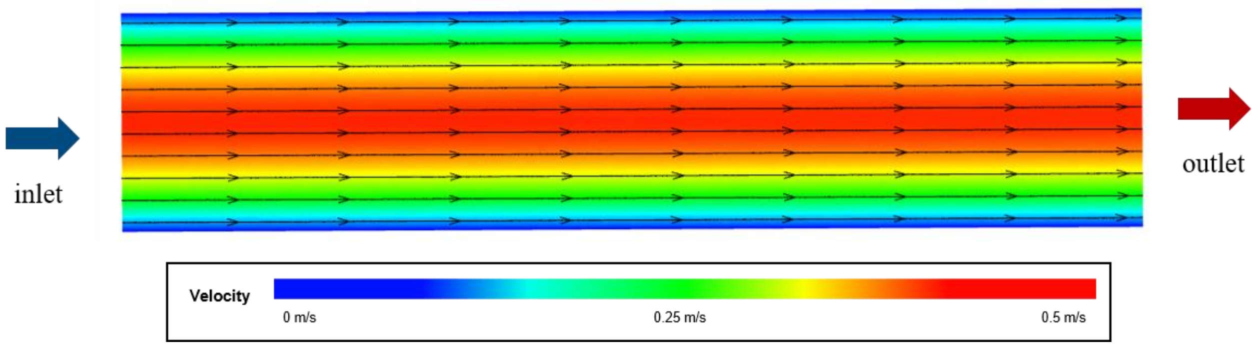
↑ Model II



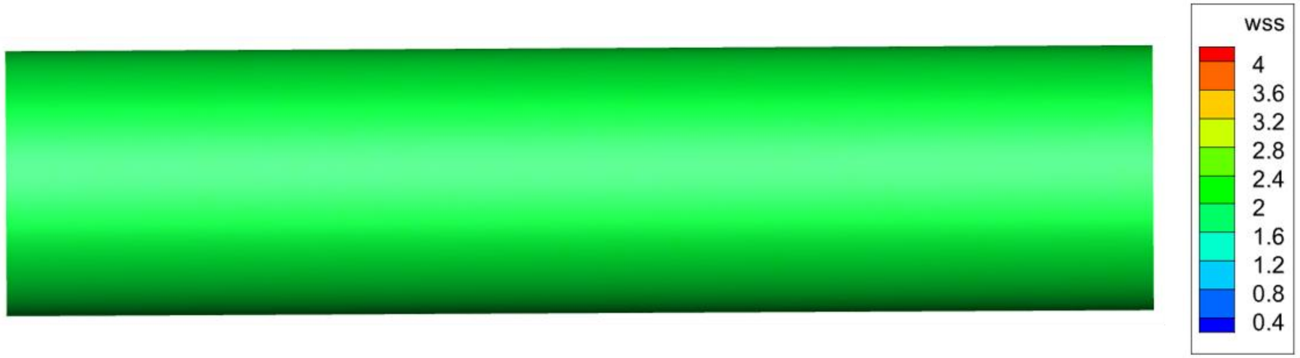
↑ Model III

Results – Model I

Streamline & Velocity



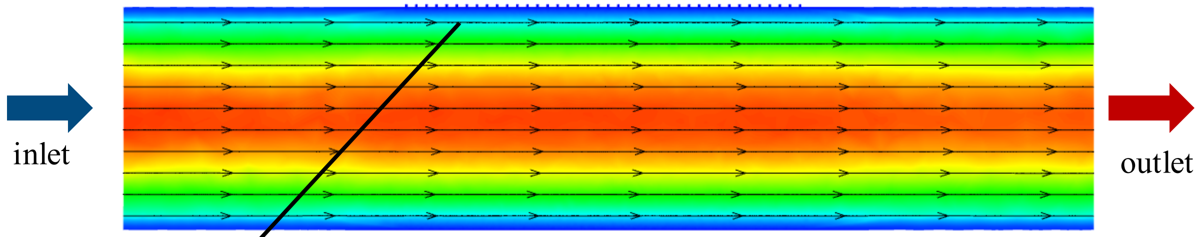
WSS Contour



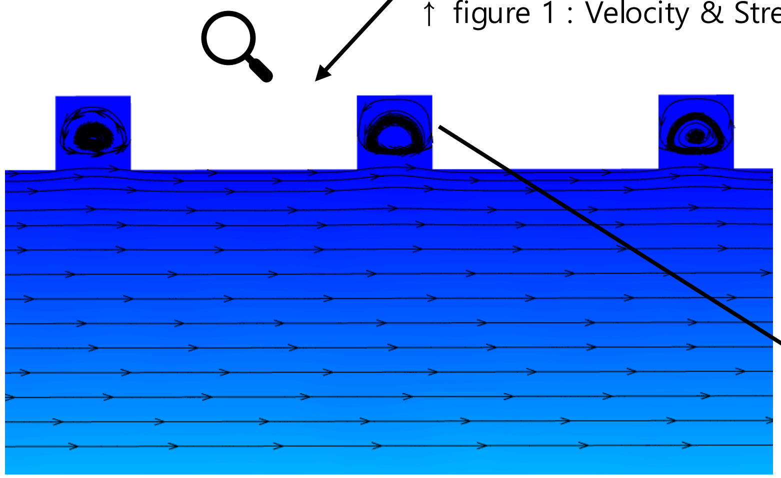
- The fully developed flow starting at the inlet is maintained, with linear streamline.
- The wall shear stress doesn't change either, staying at approximately at 2Pa.

Results – Model II

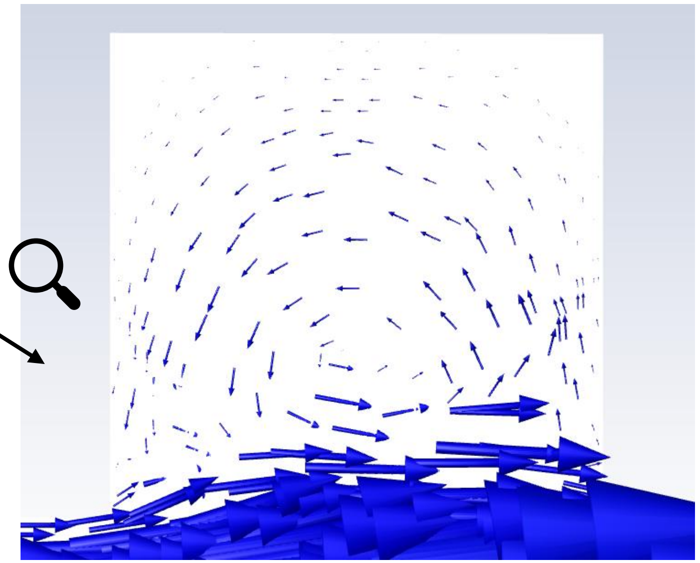
Streamline & Velocity



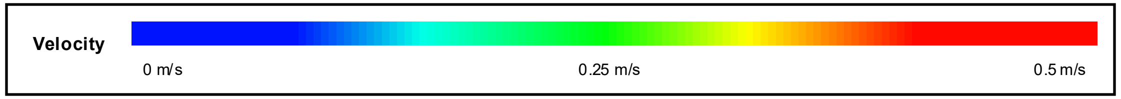
↑ figure 1 : Velocity & Streamline



↑ figure 2 : Velocity & Streamline near pattern



↑ figure 3 : velocity vector inside pattern

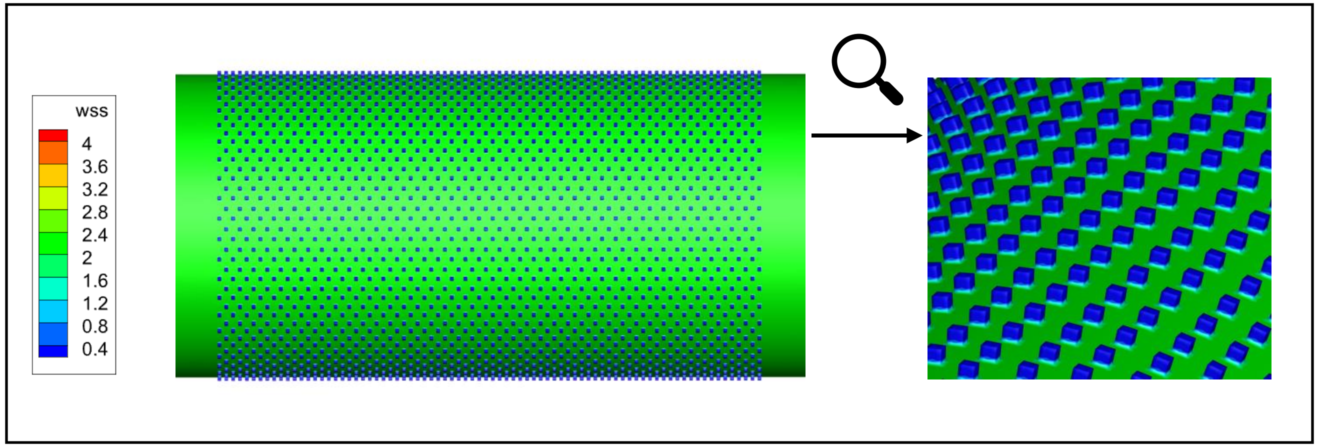


- In the aspect of mainstream, streamline and velocity was not affected. (figure 1)
- Streamline waves upwards near the pattern, and recirculation occurred inside the pattern. (figure 2)
- Looking at the velocity vector inside the pattern, we could confirm the recirculation. (figure 3)

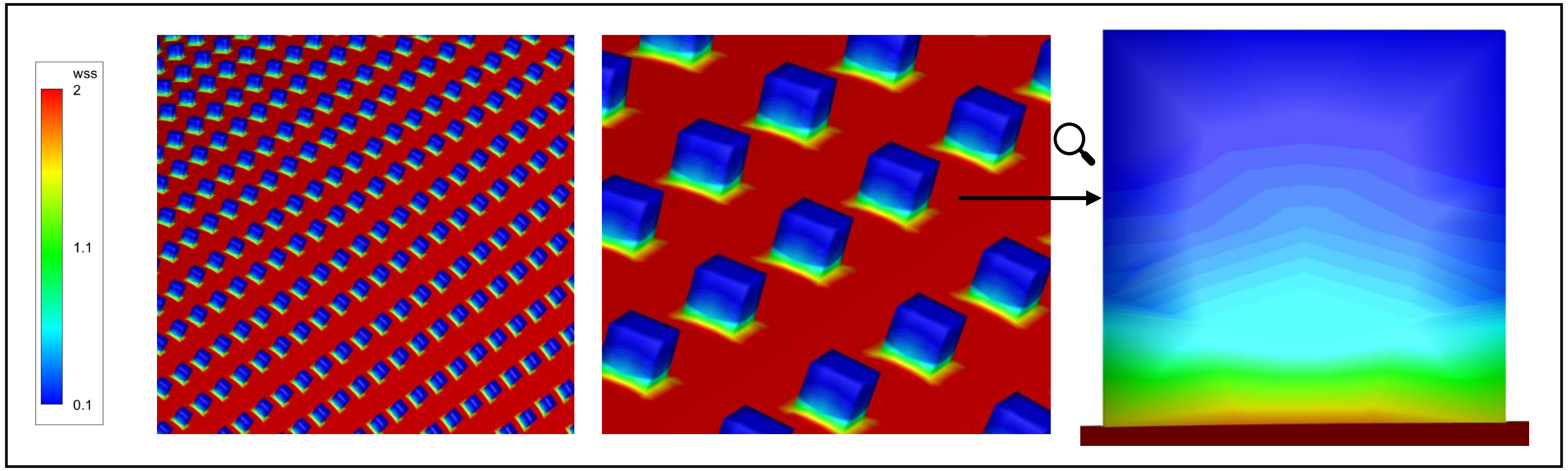
Part 2-2

Results – Model II

WSS Contour



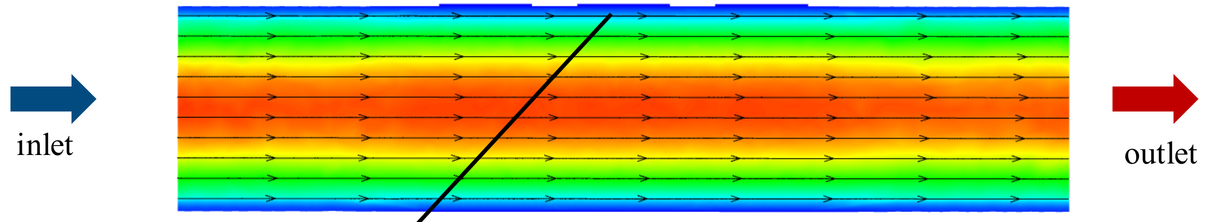
↑ figure 1 : WSS Contour (0.4 ~ 4Pa) / ↓ figure 2 : WSS Contour (0.1 ~ 2Pa)



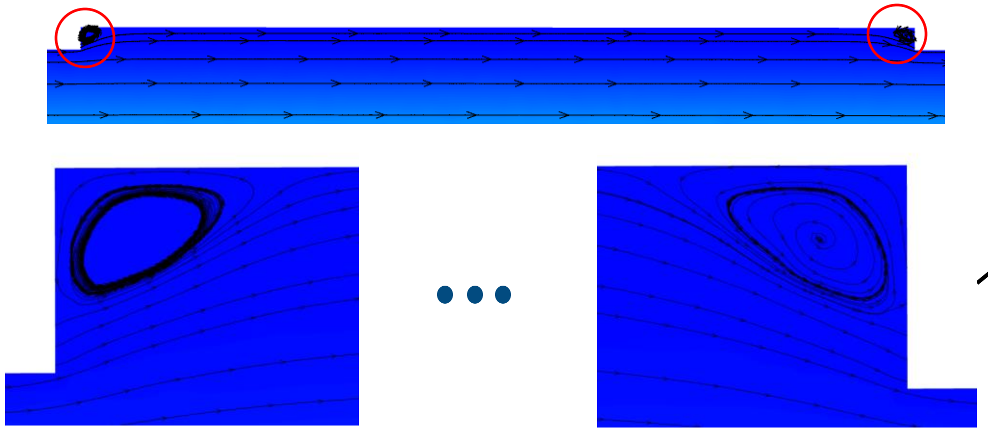
- Low wall shear stress was formed in the patterns. (figure 1)
- As it is the low wall shear stress we want to discuss, we changed the contour range to 0.1 ~ 2Pa and checked again.
- Increasing low wss was obsersved as it gets deeper inside the pattern. (figure 2)

Results – Model III

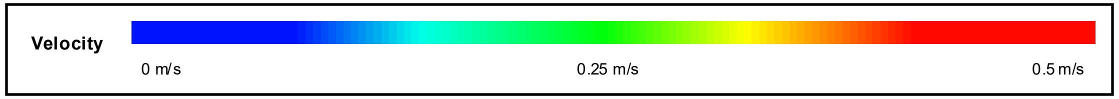
Streamline & Velocity



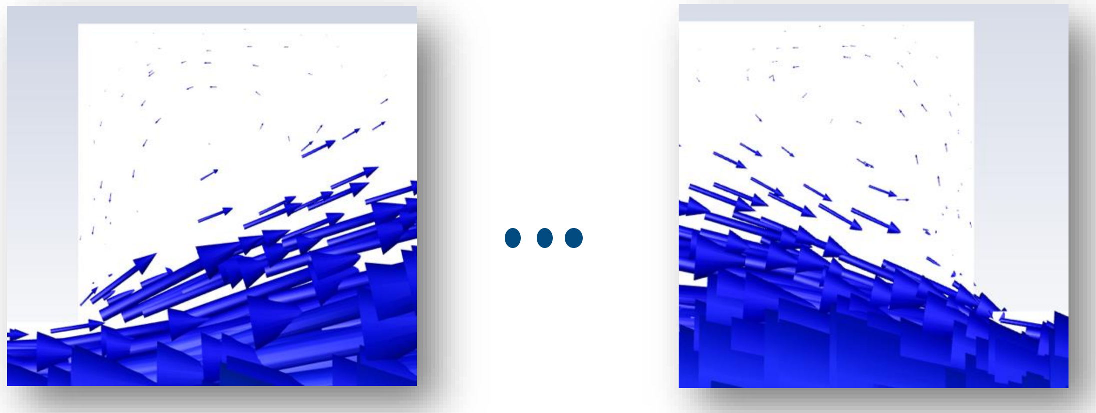
↑ figure 1 : Velocity & Streamline



↑ figure 2 : Velocity & Streamline near pattern and inside pattern



↓ figure 3 : velocity vector in the ends of the pattern

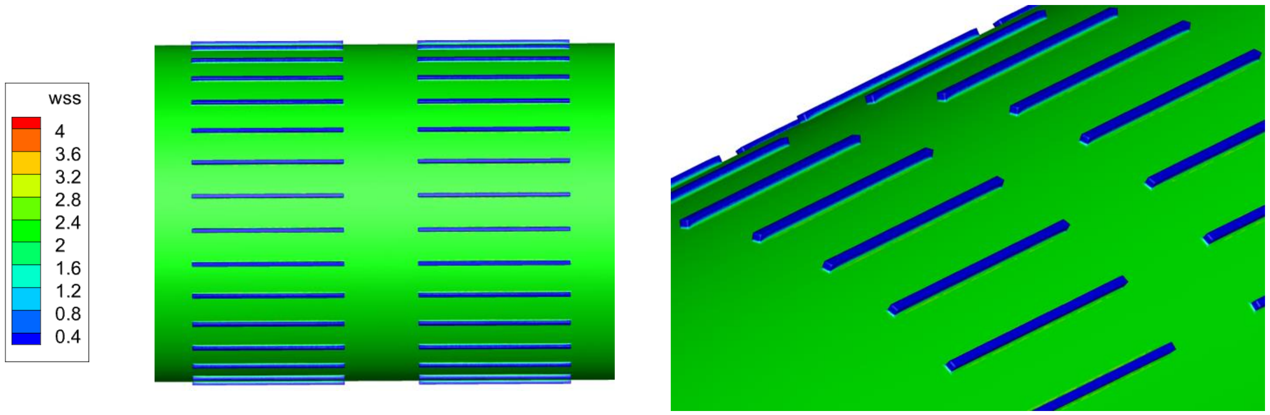


- In the aspect of mainstream, streamline and velocity was not affected. (figure 1)
- The streamline from mainstream goes through the middle part of the pattern, and recirculation is formed in the both ends of the pattern. (figure 2)
- Looking at the velocity vector inside the pattern, we could confirm the recirculation in both ends. (figure 3)

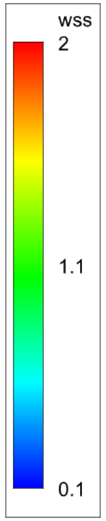
Part 2-2

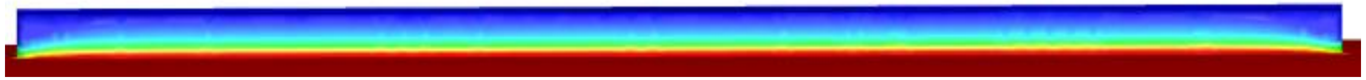
Results – Model III

WSS Contour



↑ figure 1 : WSS Contour (0.4 ~ 4Pa) / ↓ figure 2 : WSS Contour (0.1 ~ 2Pa)



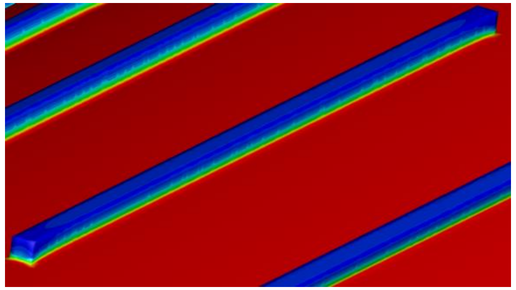


: Side view of pattern

(figure 2-1)

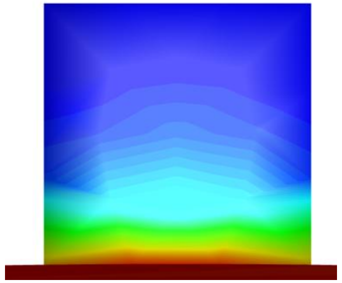
: Top view of pattern

(figure 2-2)



: isometric view of pattern

(figure 2-3)



: Front view of pattern

(figure 2-4)

- Low wall shear stress in formed in the patterns. (figure 1)
- Looking at the side view, the ends of pattern has thicker low wss region than midst of pattern. (figure 2-1)
- Looking at the top view, the wss of both ends has lower value than that of middle. (figure 2-2)
- Looking at the front view, there is lower wss in the deeper region of pattern. (figure 2-4)

Conclusion

Conclusion :

- Artificial blood vessels made from synthetic non-degradable polymers may have higher possibility of restenosis due to the recirculation and formation of low wall shear stress region caused by flow disturbance from pattern.

Reference

1) Johnson WC, Lee KK. A comparative evaluation of polytetrafluoroethylene, umbilical vein, and saphenous vein bypass grafts for femoral-popliteal above-knee revascularization: a prospective randomized Department of Veterans Affairs cooperative study. J Vasc Surg 2000;32(2):268–277. [PubMed: 10917986]

2) Trubel W, Schima H, Moritz A et al, Compliance mismatch and formation of distal anastomotic intimal hyperplasia in extremely stiffened and lumen-adapted venous grafts. Eur J Vasc Endovasc Surg 1995; 10:415-423.

3) J. Lu, M.P. Rao, N.C. MacDonald, D. Khang, T.J. Webster, Improved endothelial cell adhesion and proliferation on patterned titanium surfaces with rationally designed, micrometer to nanometer features, Acta Biomater. 4 (2008) 192– 201.

4) Reynolds MM, Hrabie JA, Oh BK, et al. Nitric oxide releasing polyurethanes with covalently linked diazeniumdiolated secondary amines. Biomacromolecules 2006;7(3):987–994. [PubMed: 16529441]

5) Kohler TR, Kirkman TR, Kraiss LW, Zierler BK, Clowes AW, Increased blood flow inhibits neointimal hyperplasia in endothelialized vascular grafts. Circ Res 1991; 69: 1557-1565.

Thanks to BFE lab

Link: BFE Lab Website

Thank You

@Kunwoo Lee